

# 1. Introduction

SalamLearn is an offline-first, interactive mobile educational application designed to augment the traditional pedagogical methods of the Philippine Madrasah system. The architecture strictly aligns with the Department of Education's (DepEd) Refined Elementary Madrasah Curriculum (ALIVE) for Grade 1 learners. To address localized constraints such as intermittent internet connectivity, the restriction of devices in the classroom to only the educator, and shared device usage at home, the app relies on a local caching framework (Hive) alongside a unified codebase (Flutter).

## 2. User Roles (Actors)

The system recognizes three distinct actors with different access privileges:

- **The Learner (Child):** Has restricted access. Can only interact with the Student Hub and Core Educational Modules. Cannot access settings, exit the app, or view external links. The Learner's 1-to-1 device interaction occurs exclusively outside of the classroom (e.g., at home).
- **The Parent/Guardian (Admin):** Requires a 4-digit PIN. Has full Create, Read, Update, Delete (CRUD) rights over the child's profile, data synchronization, and privacy consents.
- **The Asatidz (Teacher):** Requires a 4-digit PIN. Has administrative privileges to create class sections, enroll students via invitation codes, view detailed student/class progress dashboards, assign specific game modules as take-home assignments, and utilize presentation-level overrides for live classroom casting. The Teacher is the sole operator of the mobile device during live physical classroom sessions.

## 3. Functional Requirements (FRs)

### Module 1: Onboarding & System Initialization

- **FR-1.1:** The system shall execute a background script upon launch to query the local database and verify the integrity of all critical offline assets (.mp3, .png, .json).
- **FR-1.2:** The system shall validate the host device's available local storage and halt initialization if the available storage is less than 500MB, triggering a high-priority UI alert.
- **FR-1.3:** The system shall provide a UI toggle to set the primary instructional language (English or Filipino) for all non-Arabic interface text.

### Module 2: Authentication & Profile Management

- **FR-2.1:** The system shall support dual-state authentication: Firebase Auth for online states and locally hashed credentials stored via Hive for offline validation.
- **FR-2.2:** The system shall present a mandatory consent UI to log a timestamp confirming the parent agrees to local telemetry collection under the Philippine Data Privacy Act of 2012 (RA 10173).
- **FR-2.3:** The system shall allow parents to create a Learner entity and generate an encrypted 4-digit PIN to lock administrative and teacher modules.

### Module 3: Student Hub & Rewards

- **FR-3.1:** The system shall present a text-minimal, high-contrast dashboard with large touch targets tailored for children ages 5 to 11.
- **FR-3.2:** The system shall display a visual "Streak Tracker" indicating consecutive days played to encourage routine engagement.

- **FR-3.3:** The system shall populate a "Digital Backpack" with achievement badges triggered by specific Grade 1 module completion milestones.

#### **Module 4: Core Educational Modules (Wag muna toh, di pa finalize)**

- **FR-4.1:** The system shall provide a Tactile Recognition Engine that tracks X/Y coordinate inputs and validates user continuous drag events against predefined right-to-left vector paths for Arabic letters and numerals.
- **FR-4.2:** The system shall provide an offline Pronunciation Audio Mapper that triggers native media playback upon tap gestures on digital flashcards.
- **FR-4.3:** The system shall feature a Media Playback Controller for Qur'an/Hadith that highlights text strings on the screen synchronously with audio timestamps.
- **FR-4.4:** The system shall render a static 2D storyboard progression for Sirah/Aqidah topics using state-based view switching.
- **FR-4.5:** The system shall execute a Collision & Array Engine for Fiqh/Taharah (e.g., Wudhu steps, Halal/Haram) that validates if a dragged item ID matches a target drop-zone ID.

#### **Module 5: Parent Analytics Dashboard**

- **FR-5.1:** The system shall process raw local gameplay telemetry (stroke accuracy percentage, sequencing errors, time-on-task) and render it into visual charts.
- **FR-5.2:** The system shall generate automated, curriculum-aligned text prompts suggesting physical practice or active mediation based on the child's recorded weak points.

#### **Module 6: Teacher & Offline Classroom Mode (1-to-Many Architecture)**

- **FR-6.1 (Class Management):** The system shall allow verified teachers to create and manage multiple class sections, generate invitation codes, and enroll students/parents into their respective classes.
- **FR-6.2 (Teacher Analytics):** The system shall provide a Teacher Dashboard that aggregates telemetry data to display learner completion rates, tracing accuracy, lesson mastery, and recent activity for all enrolled students.
- **FR-6.3 (Take-Home Assignments):** The system shall enable teachers to assign specific (previous or advanced) learning modules to an entire class or specific students, which will appear in the Learner's "home mode" for independent practice.
- **FR-6.4 (Screen Casting):** The system shall automatically scale UI components and force a 16:9 landscape orientation when the Presentation UI is cast to an external projector or TV, guaranteeing that interactive lessons can be delivered to the whole class using only the teacher's device, operating under a strict zero-student-device classroom model.
- **FR-6.5 (Progression Override):** The system shall allow the Teacher account to suspend Learner save-state data, granting instant progression lock overrides for presentation purposes.
- **FR-6.6 (Hot Seat):** The system shall provide a "Hot Seat" mode enabling the teacher to project tracing or sorting modules while a single student volunteer comes to the front of the classroom to solve the sequence directly on the teacher's host device.
- **FR-6.7 (Choral Controller):** The system shall provide a Choral Reading controller for the teacher to trigger audio files without game progression locks for whole-class repetition (Talqeen).

#### **Module 7: Settings & Data Lifecycle**

- **FR-7.1:** The system shall provide independent volume control sliders for background audio, UI sound effects, and pronunciation voices.
- **FR-7.2:** The system shall feature a manual synchronization trigger to push local offline

data to the cloud when Wi-Fi is detected.

- **FR-7.3:** The system shall execute a local data erasure protocol (Hive.deleteFromDisk()) to satisfy the "Right to be Forgotten" constraint.

#### 4. Non-Functional Requirements (NFRs)

| NFR Category | Requirement Description                                                                                                                            |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Performance  | The system shall maintain 60 to 120 FPS rendering during tactile tracing via Flutter's Ahead-of-Time compilation to prevent visual stuttering.     |
| Storage      | The system shall require a minimum of 500MB of local device storage to cache core educational assets independently of cloud connectivity.          |
| Security     | The system shall hash user credentials locally and encrypt the 4-digit PIN to restrict unauthorized access to administrative modules.              |
| Legal        | The system shall restrict telemetry collection from minors without explicit, logged parental consent to comply with RA 10173.                      |
| Usability    | The system shall adhere to ISO/IEC 25010 child HCI guidelines by utilizing single-finger gestures and avoiding complex multi-finger shortcuts.     |
| Reliability  | The system shall execute offline-first reads/writes via a local NoSQL database (Hive) to ensure zero data loss during intermittent network states. |

#### 5. Architectural Data Flow & System State Diagrams (Textual)

##### System Conceptual Architecture (IPO Model)

- **Input:** DepEd ALIVE Curriculum rules, Child HCI UI constraints (large targets, high contrast), RA 10173 compliance mandates, offline 500MB assets, Teacher assignments via cloud sync.
- **Process:** Flutter AOT Compilation rendering, Hive local caching scripts, User input validation via Collision/Tactile engines, Background cloud-sync logic for assignments and telemetry.
- **Output:** Evaluated offline modules (Tracing, Sorting, Casting), push-notification/UI updates for teacher-assigned homework, and telemetry charting in the Parent and Teacher Dashboards.

##### Data Flow Sequence (Learner to Dashboard)

1. **Classroom Flow (Single Device):** (*UPDATED*) The Teacher casts their device to the classroom display. The Teacher acts as the primary input operator (or allows one volunteer via Hot Seat) while the class observes. No learner telemetry is generated or logged during this presentation state.
2. **Assignment (Cloud to Local):** The Teacher assigns a module via their dashboard for

home practice. Upon Wi-Fi detection, the child's device syncs and flags the specific module as "Assigned" in the Student Hub.

3. **Home Interaction (1-to-1 Device): (UPDATED) The Learner operates the app outside the classroom on a parent's device.** Learner performs a drag-and-drop action or tracing event on the device screen during "home mode".
4. **Validation:** Flutter UI sends X/Y coordinates to the Collision Engine.
5. **Local Storage:** The engine validates the placement, generates telemetry (time-on-task, error count), and writes synchronously to the local **ProgressBox** in the Hive database.
6. **Retrieval (Offline - Parent):** When the Parent opens the Analytics Dashboard, the system queries the Hive database and renders a local chart mapping the errors.
7. **Cloud Sync (Asynchronous - Teacher):** Upon detecting Wi-Fi, the system's background **SyncManager** reads the **ProgressBox** and pushes the telemetry data to Firebase, allowing the Teacher to view the child's mastery of the assigned take-home modules from their own device.